

1.4 The matrix equation $A\vec{x} = \vec{b}$

I

$$A\vec{x} = [\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n$$

By this definition $A\vec{x}$ is the linear combination of the columns of A using the corresponding entries in $\vec{x}$ as weights.

If A is an $m \times n$ matrix with columns $\vec{a}_1 \dots \vec{a}_n$, and if $\vec{b}$ is in $\mathbb{R}^m$, the matrix equation

$A\vec{x} = \vec{b}$ has the same solution set as the vector equation $x_1\vec{a}_1 + \dots + x_n\vec{a}_n = \vec{b}$.

Which in turn has the same solution set as the system of linear equations whose augmented matrix $[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n | \vec{b}]$

These definitions leads to the fact that $A\vec{x} = \vec{b}$ iff $\vec{b}$ is a linear combination of the columns of A .

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A , either they are all true statements or they are all false.

- For each $\vec{b}$ in $\mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a solution.
- Each $\vec{b}$ in $\mathbb{R}^m$ is a linear combinations of the columns of A .
- The columns of A span $\mathbb{R}^m$.
- A has a pivot in every row.

A is a coefficient matrix

1.4 The Matrix Equation $A\vec{x} = \vec{b}$ (cont'd)

II

Definition

If A is an $m \times n$ matrix, with columns $\vec{a}_1, \dots, \vec{a}_n$, and if $\vec{x}$ is in $\mathbb{R}^n$, then the product of A and $\vec{x}$, denoted by $A\vec{x}$, is the linear combination of the columns of A using the corresponding entries in $\vec{x}$ as weights; that is,

$$A\vec{x} = [\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n$$

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Note that $A\vec{x}$ is defined only if the number of columns of A equals the number of entries in $\vec{x}$.

Theorem 3

If A is an $m \times n$ matrix, with columns $\vec{a}_1, \dots, \vec{a}_n$, and if $\vec{b}$ is in $\mathbb{R}^m$, the matrix equation

$$A\vec{x} = \vec{b} \quad (4)$$

has the same solution set as the vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n = \vec{b} \quad (5)$$

which, in turn, has the same solution set as the system of linear equations whose augmented matrix is

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n \mid \vec{b}] \quad (6)$$

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1.4 The Matrix Equation $A\vec{x} = \vec{b}$ (cont'd)

III

The equation $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is a linear combination of the columns of A .

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Example 3

$$\text{Let } A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Is the equation $A\vec{x} = \vec{b}$ consistent for all possible values of b_1, b_2, b_3 ?

Solution

Let's row reduce the augmented matrix for $A\vec{x} = \vec{b}$ and find out.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{array} \right]$$

$$R_2 \leftarrow R_2 + 4R_1$$

$$R_3 \leftarrow R_3 + 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 7 & 5 & b_3 + 3b_1 \end{array} \right]$$

$$R_2 \leftarrow R_2 \cdot \frac{1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 7 & 5 & \frac{1}{2}(b_2 + 4b_1) \\ 0 & 7 & 5 & b_3 + 3b_1 \end{array} \right]$$

$$R_3 \leftarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 7 & 5 & \frac{1}{2}(b_2 + 4b_1) \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{array} \right]$$

cleaning up

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 7 & 5 & \frac{1}{2}b_2 + 2b_1 \\ 0 & 0 & 0 & b_1 - \frac{1}{2}b_2 + b_3 \end{array} \right]$$

The third entry in column 4 equals $b_1 - \frac{1}{2}b_2 + b_3$. The equation $A\vec{x} = \vec{b}$ is not consistent for every $\vec{b}$, because some choices of $\vec{b}$ can make $b_1 - \frac{1}{2}b_2 + b_3$ nonzero.

1.4 The Matrix Equation $A\vec{x} = \vec{b}$ (cont'd)

In the next theorem, the sentence "The columns of A span $\mathbb{R}^m$ " means that every $\vec{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A . In general, a set of vectors $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^m$ spans (or generates) $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_p$ - that is, if $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\} = \mathbb{R}^m$.

Theorem 4

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A , they are either all true or all false.

- For each $\vec{b}$ in $\mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a solution.
- Each $\vec{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A .
- The columns of A span $\mathbb{R}^m$.
- A has a pivot position in every row.

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IV

Computation of $A\vec{x}$

Example 4

Compute $A\vec{x}$, where $A = \begin{matrix} & \vec{a}_1 & \vec{a}_2 & \vec{a}_3 \\ \begin{matrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{matrix} & \begin{matrix} 3 \times 3 \\ m \times n \end{matrix} \end{matrix}$ and $\vec{x} = \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} \begin{matrix} 3 \times 1 \\ n \times p \end{matrix}$

Solution

Using theorem 4 b. we can write

$$\begin{aligned} A\vec{x} &= \vec{x}_1 \vec{a}_1 + \vec{x}_2 \vec{a}_2 + \vec{x}_3 \vec{a}_3 \\ &= \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ -1x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{bmatrix} \\ &\quad \begin{matrix} 3 \times 1 \\ m \times p \end{matrix} \end{aligned}$$

Row-Vector rule for computing $A\vec{x}$

If the product $A\vec{x}$ is defined, then the i th entry in $A\vec{x}$ is the sum of the products of corresponding entries from row i of A and from the vector $\vec{x}$.

1.4 The Matrix Equation $A\vec{x} = \vec{b}$ (cont'd)

V

Example 5

$$a. \begin{matrix} & 2 \times 3 & 3 \times 1 & & 2 \times 1 \\ \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} & \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} & = & \begin{bmatrix} 1 \cdot 4 + 2 \cdot 3 - 1 \cdot 7 \\ 0 \cdot 4 - 5 \cdot 3 + 3 \cdot 7 \end{bmatrix} & = & \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{matrix}$$

$$b. \begin{matrix} & 3 \times 2 & 2 \times 1 & & 3 \times 1 \\ & m \times n & n \times p & & m \times p \\ \begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} & \begin{bmatrix} 4 \\ 7 \end{bmatrix} & = & \begin{bmatrix} 2 \cdot 4 - 3 \cdot 7 \\ 8 \cdot 4 + 0 \cdot 7 \\ -5 \cdot 4 + 2 \cdot 7 \end{bmatrix} & = & \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix} \end{matrix}$$

$$c. \begin{matrix} & 3 \times 3 & 3 \times 1 & & 3 \times 1 \\ & m \times n & n \times p & & m \times p \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} r \\ s \\ t \end{bmatrix} & = & \begin{bmatrix} r \\ s \\ t \end{bmatrix} \end{matrix}$$

Proof

$$n=3, A = [\vec{a}_1 \ \vec{a}_2 \ \vec{a}_3], \vec{u}, \vec{v} \in \mathbb{R}^3$$

$$A(\vec{u} + \vec{v}) = [\vec{a}_1 \ \vec{a}_2 \ \vec{a}_3] \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}$$

$$= \underbrace{\vec{a}_1(u_1 + v_1) + \vec{a}_2(u_2 + v_2) + \vec{a}_3(u_3 + v_3)}_{\substack{\vec{u} + \vec{v} \\ \text{columns of } A}} \\ = (\vec{a}_1 u_1 + \vec{a}_2 u_2 + \vec{a}_3 u_3) + (\vec{a}_1 v_1 + \vec{a}_2 v_2 + \vec{a}_3 v_3)$$

$$= A\vec{u} + A\vec{v}$$

$$A(c\vec{u}) = [\vec{a}_1 \ \vec{a}_2 \ \vec{a}_3] \begin{bmatrix} c u_1 \\ c u_2 \\ c u_3 \end{bmatrix} = (c u_1) \vec{a}_1 + (c u_2) \vec{a}_2 + (c u_3) \vec{a}_3$$

$$= c(u_1 \vec{a}_1) + c(u_2 \vec{a}_2) + c(u_3 \vec{a}_3)$$

$$= c(A\vec{u})$$

Theorem 5

If A is an $m \times n$ matrix, $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then:

$$a. A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$$

$$b. A(c\vec{u}) = c(A\vec{u})$$